

Fleet Performance and Efficiency Dashboard

Power BI Module-End Assignment — Dashboard Documentation

1. Overview

This document summarizes the Power BI report contained in `powerBI_module_end_assignment.pbix`. The report is a single-page fleet management dashboard titled “Fleet Performance and Efficiency Dashboard in Power BI”, built to track vehicle trips, delivery performance, fuel efficiency, and maintenance cost.

2. Objectives

This dashboard was built to give a fleet manager a single view to:

- Monitor overall fleet health — track total trips, average distance per trip, and average cost per km at a glance via top-line KPI cards.
- Measure delivery reliability — surface on-time delivery % overall and broken down by destination, to spot problem routes.
- Diagnose the drivers of poor delivery performance — use the Key Drivers visual to identify which factors (vehicle type, driver, distance) most affect delivery status, enabling root-cause investigation rather than just reporting the symptom.
- Track efficiency over time — visualize fuel efficiency trends by date to catch gradual degradation or improvement.
- Control maintenance and running cost — compare maintenance cost across vehicle types and drill into cost-per-km through an interactive decomposition tree, down to the vehicle-type / distance level.
- Support ad-hoc filtering — let the user slice the whole page by vehicle type without switching reports.

In short: move from “what happened” (KPIs) to “why it happened” (Key Drivers, decomposition tree) on a single page.

3. Key Insights This Dashboard Is Designed to Surface

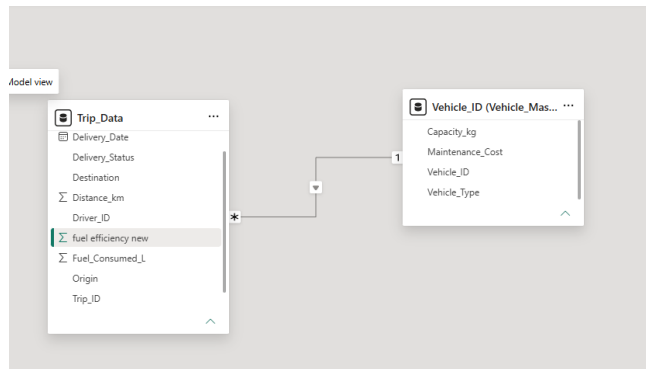
These are the questions the report structure is built to answer once populated with live data — refresh the model in Power BI Desktop to see the actual current figures.

- Cost efficiency: whether cost-per-km is being driven more by fuel/distance or by maintenance, and which vehicle type is the most expensive to run.
- Delivery reliability: which destinations have the worst on-time delivery %, and whether that's tied to a specific vehicle type or driver rather than distance alone.
- Root cause of delays: the Key Drivers visual ranks the factors most associated with a poor delivery status, so you can act on the biggest lever instead of guessing.
- Fleet aging / efficiency drift: the fuel-efficiency-by-date trend line makes it easy to spot whether efficiency is declining over time (a signal for maintenance or fleet renewal).
- Cost concentration: the decomposition tree lets you drill from total cost-per-km down into the vehicle types and distance bands that contribute most, rather than only seeing an average.

4. Data Model

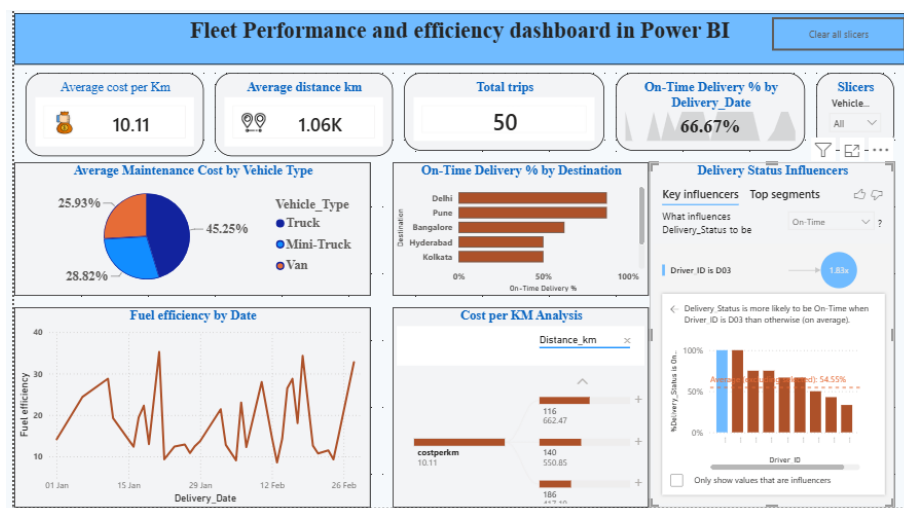
The report's data model contains the following tables and relationships (from the model diagram layout):

- Trip_Data — fact table holding individual trips (Trip_ID, Delivery_Date, Delivery_Status, Destination, Driver_ID, Distance_km, fuel efficiency new)
- Vehicle_Master — dimension table with vehicle attributes (Vehicle_ID, Vehicle_Type, Maintenance_Cost)
- Table — a measures table (On-Time Delivery %, costperkm, Total Trips)
- Trip_Data relates to Vehicle_Master via Vehicle_ID



5. Page Layout Schematic

The diagram below reproduces the exact position and size of every visual on the report page, taken directly from the report layout definition (page canvas 1280 × 720).



6. Visual Inventory

Each visual on the dashboard, its purpose, and the underlying fields or measures it uses:

Visual Type	Title	Fields / Measures Used
Card	Average cost per Km	Table.costperkm
Card	Average distance km	Sum(Trip_Data.Distance_km)
Card	Total trips	Table.Total Trips

KPI	On-Time Delivery %	Table.On-Time Delivery %, Trip_Data.Delivery_Date
Clustered Bar Chart	On-Time Delivery % by Destination	Trip_Data.Destination, Table.On-Time Delivery %
Pie Chart	Average Maintenance Cost by Vehicle Type	Vehicle_Master.Vehicle_Type, CountNonNull(Vehicle_Master.Maintenance_Cost)
Line Chart	Fuel efficiency by Date	Sum(Trip_Data.fuel efficiency new), Trip_Data.Delivery_Date
Decomposition Tree	Cost per KM Analysis	Table.costperkm, Trip_Data.Distance_km, Vehicle_Master.Maintenance_Cost, Vehicle_Master.Vehicle_Type
Key Drivers	Delivery Status Influencers	Trip_Data.Delivery_Status, Vehicle_Master.Vehicle_Type, Trip_Data.Driver_ID, Sum(Trip_Data.Distance_km)
Slicer	Vehicle Type filter	Vehicle_Master.Vehicle_Type
Button	Reset / navigation action	—

7. Dashboard Narrative

The top row of KPI cards gives an at-a-glance summary: average cost per kilometre, average distance travelled, total trips, and an on-time delivery % KPI.

The middle row breaks this down further — a pie chart shows how average maintenance cost varies by vehicle type, while a clustered bar chart shows on-time delivery % by destination.

The right-hand key drivers visual surfaces which factors (vehicle type, driver, distance) most influence delivery status, supporting root-cause analysis.

The bottom row adds a fuel-efficiency trend line over time and a decomposition tree that lets a user drill into cost-per-km by vehicle type and distance, for interactive root-cause exploration.

A slicer (vehicle type) and an action button (likely a reset/navigation control) round out the interactive elements of the page.

8. Recommendation

To complete this as a submission-ready assignment file, open the .pbix in Power BI Desktop, confirm the visuals render as expected with your data, and take screenshots of the full dashboard (and any tooltip/drill-through states you want to showcase). Share those images and they can be inserted directly into this document, replacing or supplementing the schematic above.